INSTITUTO SUPERIOR TÉCNICO

Departamento de Engenharia Informática

Forensics Cyber Security

MEIC / METI 2025-2026 – 1st Period

Digital Forensics Report Lab3

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1. Acquired artifacts

Name	Type	SHA-256 Value
irc_chat_1	TXT	a449263a629e157dbbd60c753b93cb49fa543f05051c543925edb7cd9684e97c
irc_chat_2	TXT	256a46def583ac7861ed46755941d1eeb91b2741a45b5cf78bd2a838b8366175
irc_chat_3	TXT	82cfe3d44716c3cc4da4ac1615f26972af2789672d29e2eb9a1c4b829cf0a18b
normalized.txt	TXT	69ed8e73d0874169f52e8e69eed928e29a82a439cc3d12aa28ea9c8b3ed6a489
nyinuq.png	PNG	ff56d339f5affe0279530214dec0ef01950828beb4af6bfa83fe8465a71da7d8
irc_chat_4	TXT	1a784c9fe8d3749327c8e5af60c9687ab7ba0eb35dbe983d937a5fab4c9fd021
ftp_transfer	TXT	63946e4efc85fabf9a4f6f95e8144577667454c4df073d0be900318d613a6b78
parking.db	SQLITE3 DB	23108d6552bbd633ed80c3de50449e0698294c0da3a049a61bbaed41e0edcea0
adelino.eml	EML	49b2145a07148626cfd58527947eddecd332996c7740283450d0a14f599ec904
Re Update for the Smart Algorithm - Duarte Calado.eml	EML	043dc8284a24e3cfaea036259569261079dad68c5b40a0689715cabd5d586481
Tutorial-short%20%28English%29.pdf	PDF	d70c1cc772bd3b8251cf64de4b5aee871f3200ee4b8e7870a78718a25d4ee00c

irc_chat_5	TXT	15c670ed081724b8dc12e268d1c480b7d701d2d57ed30b5744b9bab0e1b35a78
irc_chat_6	TXT	fa92c463f98c14b3e49ce7f0ecbf08a6a5d18a949b5b4201af66cbf0cc29c2d6
jygyt1.png	PNG	c938a05b84f2d749fc6ac65a88f57ccc0c860d4169577947425cc01aa1bbd1c3
file1.txt	TXT	14a8739a799f0c17d665f555fc96f8629b75fc3bfb4f74272d766b4c269c1ca3
file2.txt	TXT	1293754e8fcfaba1979d4709d433580d78ee5cb5181bb1cf0502967114524b8a
Miguel/phishing_email_Miguel.json	JSON	07aaadff6c2e67dc8bab05c52d4056babb9e36e292ec2c328b0a093d5c8ad61d
Miguel/parsed_headers.json	JSON	7f08aef6c6f2333e6a10711877d86888ac240445c98ec6395dd924449cde258a
Miguel/attachments.json	JSON	65cb06f21f1c3afb299034adcb72bf70687a4c75d3a258c4de281455c4096c19
Miguel/email_summary.txt	TXT	e43361c43c8c176f685d9a999fa173500c3f86ed975ed0a14731d932c76539ba
Miguel/message.pgp.asc	ASC	baae6c16e781c8ee00dcb86d881045300b2c00102d369b92d98944f78afb470d
Adelino/phishing_email_Adelino_response.json	JSON	97cedd9734e7d457df3311261edac587e44071502b92989b397248fbeff2e72d
Adelino/parsed_headers.json	JSON	84f616e44a281d36e86edfee01ba3b3c3f40630f2680184a603c6dc90fc1b9c5
Adelino/email_summary.txt	TXT	b5759312197cd7df7fe26ec601f1e7d92ac1d690a536ede2e022d2f0cfaead25
Adelino/message.pgp.asc	ASC	3ed8cd4e1063e27baa32a2e48e877dc691e34c95c728613d953f926ba4e02bb1

2. Report of all findings

The third stage of the MEMELgate investigation focused on consolidating and expanding the evidence gathered in the previous phases, with particular emphasis on the fraudulent "smart parking" scheme and its operational and financial mechanisms. Acting under the guidance of the Polícia Judiciária and with the support of Miguel Pontes, an intern from MEMEL's gateway security division, the investigation team was provided with access to relevant forensic material, including network logs, from MEMEL's internal systems. The objective of this stage was to perform a comprehensive analysis of these artifacts, verify their authenticity, and map the interactions between internal staff and external actors to fully reconstruct the scope and structure of the ongoing fraudulent activity. For the analysis of network traffic, Wireshark was employed to examine packet captures in detail, enabling the identification of communication streams, protocol usage, and connections relevant to the investigation.

Trace 1:

The analysis began with the first network trace (Trace 1). Early in the capture, at packet 15, a DNS query was observed from Miguel's workstation (IP 194.210.61136) targeting `irc.freenode.net`. Shortly afterwards, at packet 21, TCP stream 0 was identified, corresponding to the first IRC chat between Miguel and an individual later confirmed to be Catarina Pato. Upon recognizing that this communication used the IRC protocol, a targeted search was performed using Wireshark's display filter for `IRC` traffic, which allowed the identification of two additional streams of interest, resulting in a total of three streams analyzed:

- **tcp.stream eq 0:** Start packet 21 – End packet 19655;
- **tcp.stream eq 128:** Start packet 19668 – End packet 19969;
- **tcp.stream eq 130:** Start packet 19994 – End packet 20465;

Within the final stream (tcp.stream eq 130), several lines of code were observed, which were later identified as corresponding to the secret previously recovered as a diff. In addition, Miguel shared content from Reddit, including the URL *https://files.catbox.moe/nyinuq.png*, which corresponded to another previously identified secret (nyinuq.png).

Other notable DNS activity was observed in the main capture:

- Packet 261 - Access to Spotify via Google DNS: open.spotify.com;
- Packet 976 - DNS query to www.google.com;

This preliminary analysis established the foundation for identifying IRC communications and subsequent interactions relevant to the fraudulent scheme.

Relevant packets:

Packet No.	Source IP	Destination	Protocol	Description
15	194.210.61136 (Miguel)	chat.freenode.net	DNS	Initial DNS query to irc.freenode.net
21	194.210.61136 (Miguel)	chat.freenode.net	TCP	Start of TCP stream 0 (IRC chat with Catarina Pato)
19655	194.210.61136 (Miguel)	chat.freenode.net	TCP	End of TCP stream 0
19668	194.210.61136 (Miguel)	chat.freenode.net	TCP	Start of TCP stream 128
19969	194.210.61136 (Miguel)	chat.freenode.net	TCP	End of TCP stream 128
19994	194.210.61136 (Miguel)	chat.freenode.net	TCP	Start of TCP stream 130
20465	194.210.61136 (Miguel)	chat.freenode.net	TCP	End of TCP stream 130
261	194.210.61136 (Miguel)	dns.google	DNS	Access to Spotify via open.spotify.com
976	194.210.61136 (Miguel)	dns.google	DNS	Standard query to www.google.com

Trace 2:

The second network trace was analysed using the same approach as Trace 1. After detecting IRC traffic, an IRC display filter was applied and an IRC conversation of interest was recovered:

- **tcp.stream eq 1620:** Start packet 91054 – End packet 91929;

In addition to IRC analysis, HTTP objects were enumerated via **File → Export Objects → HTTP**, producing a list of webpages and resources visited by the host. After enumerating the HTTP objects, each listed item was verified by opening its associated packet and applying **Follow → HTTP Stream** and the exact URL was shown in the GET request (with the exception of the KTH tutorial, which was exported directly as a PDF from the HTTP objects). The following table summarises the most relevant HTTP objects recovered and the associated packet/stream references.

Website / Resource	Packet No.	Notes
https://www.kali.org/tools/arp-scan/	4467	ARP scanning - research
https://nmap.org/book/man-port-scanning-techniques.html	16706	Nmap port-scanning techniques - research
https://nmap.org/book/man-port-specification.html	17156	Nmap port specification - research
https://www.geeksforgeeks.org/ethical-hacking/port-scanning-techniques-by-using-nmap/	18239	Tutorial on port scanning with Nmap - investigation research
https://belocal.pt/en/parking-in-lisbon/	67793	Local parking information - domain/area research relevant to scheme
https://en.wikipedia.org/wiki/Smart_city	79985	Conceptual background on "smart city" deployments
https://futurointeligente.pt/	85947	Direct domain of Futuro Inteligente
https://www.kali.org/tools/medusa	62946	Brute-force/credential tool - investigation of attack tools
https://www.youtube.com/watch?v=ifbwTt3_oCg	12904	YouTube tutorial - "How To Use nmap To Scan For Open Ports"
https://www.youtube.com/watch?v=MvZItoxsQYs	12905	YouTube video - "Port Scanning Explained"
https://www.youtube.com/watch?v/KLPzEwSypqU	12916	YouTube tutorial – "Perform a Port Scan with Nessus Professional"
https://lisboaaberta.cm-lisboa.pt/index.php/pt/lisboa-inteligente	81427	Municipal "Lisbon Smart" page - local governance/smart-city research
archive.ubuntu.com	63468	OS/package repository access - performed system updates
www.kth.se	62671	Hydra and Medusa Tutorial

Note on YouTube-related packets (i.ytimg.com): These entries correspond to the YouTube thumbnail CDN. When following the HTTP/2 stream in Wireshark (**Follow → HTTP/2 Stream**) for packets with *Host: i.ytimg.com*, the request shows *:method: GET* and a pseudo-header *:path* containing the thumbnail's location. The *:path* typically includes the video identifier as part of a subdirectory (for example */vi/<video_id>/mqdefault.jpg*). By extracting this *<video_id>* from the *:path*, it is possible to reconstruct the corresponding YouTube video URL as:

https://www.youtube.com/watch?v=<video_id>.

These URLs likely reflect only automatically loaded thumbnails (e.g., searches, homepage, embedded videos) and do not confirm that Miguel actually watched the videos.

An additional website identified among the exported HTTP objects was ChatGPT, which was accessed by Miguel. To retrieve the content of these interactions, the corresponding packets from the exported objects were selected and **Follow → HTTP Stream** was applied in Wireshark. Each stream was then examined substream by substream within the Wireshark interface. Substreams containing requests to the endpoint */backend-anon/f/conversation* were saved individually using **Save As**, and subsequently processed with the custom script `parse_gpt_conversation.py` to extract the conversation content in sequential order. Within the raw HTTP/2 payload visible after applying *Follow Stream*, the user prompts could be identified in the JSON structure following the "parts": field, as in:

*{"action":"next","messages":[{"id":"...","author":{"role":"user"},"create_time":...,"content":{"content_type":"text","parts":["**Can you create me ...***

The text following *"parts":["* corresponds to Miguel's prompt.

The captured interactions show multiple exchanges between Miguel and the ChatGPT service:

TCP Stream	HTTP/2 Stream	Last Packet No.	Sender	Content Description
1344	39	41130	Miguel	First prompt to ChatGPT
1344	39	41172	ChatGPT	Response to first prompt
1344	67	42940	Miguel	Second prompt
1344	67	42961	ChatGPT	Response to second prompt
1344	89	49537	Miguel	Third prompt
1344	89	49562	ChatGPT	Response to third prompt
1344	111	54584	Miguel	Fourth prompt
1344	111	54610	ChatGPT	Response to fourth prompt
1344	133	58182	Miguel	Fifth prompt
1344	133	60618	ChatGPT	Response to fifth prompt

By applying the display filter `http3.headers.header.value contains "search"`, several Google search queries initiated by Miguel were identified. These searches align closely with the browsing activity observed previously in the exported HTTP objects. The table below summarises the most relevant queries along with their probable destination websites:

Packet No.	Query (Google)	Probable destination(s) observed in HTTP objects
289	"how to do arp scan"	https://www.kali.org/tools/arp-scan/
10825	"how to do port scan"	https://nmap.org/book/man-port-scanning-techniques.html, https://nmap.org/book/man-port-specification.html, https://www.geeksforgeeks.org/ethical-hacking/port-scanning-techniques-by-using-nmap/ https://www.youtube.com/watch?v=ifbwTt3_oCg https://www.youtube.com/watch?v=MvZItoxsQYs https://www.youtube.com/watch?v/KLPzEwSypqU

32457	"chat"	https://chatgpt.com/
61045	"how to use medusa"	https://www.kali.org/tools/medusa, www.kth.se
65595	"parking in lisbon"	https://belocal.pt/en/parking-in-lisbon/
74579	"what is a smart city"	https://en.wikipedia.org/wiki/Smart_city, https://lisboaaberta.cm-lisboa.pt/index.php/pt/lisboa-inteligente
84481	"futuro inteligente"	https://futurointeligente.pt

Additionally, by navigating to **File → Export Objects → FTP Data**, three files were recovered, all originating from the IP address 194.210.61.134 (associated with Duarte Calado). The recovered artifacts are summarised below:

File Name	Size	Last Packet No.	Stream	Source IP	Description
parking.db	40 KB	65058	1445	194.210.61.134	SQLite database
adelino.eml	307 B	65348	1454	194.210.61.134	Email message from Adelino Estoiro
Re: Update for the Smart Algorithm – Duarte Calado	1.6 KB	65365	1455	194.210.61.134	Email thread between Duarte Calado and Eurico Trocos

Analysis of these email files, particularly the latter two, allowed for the identification of several email addresses directly involved in the fraudulent communications:

- adelino.estoiro@proton.me;
- duarte.calado@proton.me;
- caladoduarte21@gmail.com.

Additionally, by applying the display filter `ARP`, a large number of ARP requests were observed originating from Miguel's IP address. These requests repeatedly queried the network with messages of the form *"Who has [IP]? Tell [Miguel's IP]"*, targeting numerous different hosts.

arp

No.	Time	Source	Destination	Protocol	Length	Stream inde	Info
8578	2025-09-06 11:00:52.312508	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.5? Tell 194.210.61.136
8579	2025-09-06 11:00:52.313994	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.6? Tell 194.210.61.136
8580	2025-09-06 11:00:52.316972	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.7? Tell 194.210.61.136
8581	2025-09-06 11:00:52.320964	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.8? Tell 194.210.61.136
8582	2025-09-06 11:00:52.322870	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.9? Tell 194.210.61.136
8583	2025-09-06 11:00:52.323363	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.10? Tell 194.210.61.136
8584	2025-09-06 11:00:52.323994	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.11? Tell 194.210.61.136
8585	2025-09-06 11:00:52.326217	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.12? Tell 194.210.61.136
8586	2025-09-06 11:00:52.329708	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.13? Tell 194.210.61.136
8587	2025-09-06 11:00:52.332976	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.14? Tell 194.210.61.136
8588	2025-09-06 11:00:52.334970	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.15? Tell 194.210.61.136
8589	2025-09-06 11:00:52.335453	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.16? Tell 194.210.61.136
8590	2025-09-06 11:00:52.335715	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.17? Tell 194.210.61.136
8591	2025-09-06 11:00:52.338038	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.18? Tell 194.210.61.136
8592	2025-09-06 11:00:52.341507	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.19? Tell 194.210.61.136
8593	2025-09-06 11:00:52.345233	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.20? Tell 194.210.61.136
8594	2025-09-06 11:00:52.347199	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.21? Tell 194.210.61.136
8595	2025-09-06 11:00:52.348123	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.22? Tell 194.210.61.136
8596	2025-09-06 11:00:52.348124	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.23? Tell 194.210.61.136
8597	2025-09-06 11:00:52.349687	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.24? Tell 194.210.61.136
8598	2025-09-06 11:00:52.353536	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.25? Tell 194.210.61.136
8599	2025-09-06 11:00:52.357466	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.26? Tell 194.210.61.136
8600	2025-09-06 11:00:52.359611	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.27? Tell 194.210.61.136
8601	2025-09-06 11:00:52.359612	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.28? Tell 194.210.61.136
8602	2025-09-06 11:00:52.362187	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.29? Tell 194.210.61.136
8603	2025-09-06 11:00:52.362187	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.30? Tell 194.210.61.136
8604	2025-09-06 11:00:52.365920	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.31? Tell 194.210.61.136
8605	2025-09-06 11:00:52.369512	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.32? Tell 194.210.61.136
8606	2025-09-06 11:00:52.371410	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.33? Tell 194.210.61.136

Figure 1: ARP requests issued by Miguel's workstation (using resolved hostnames)

Furthermore, by applying the display filter `FTP`, numerous authentication attempts were observed, indicating that Miguel was repeatedly trying to access Duarte's FTP service, and eventually achieved "duartinhothebest", the correct password.

ftp

No.	Time	Source	Destination	Protocol	Length	Stream inde	Info
64595	2025-09-06 11:06:42.294536	Miguel Pontes	Duarte Calado	FTP	79	1428	Request: USER duarte
64597	2025-09-06 11:06:42.294536	Duarte Calado	Miguel Pontes	FTP	100	1428	Response: 331 Please specify the password.
64598	2025-09-06 11:06:42.294677	Miguel Pontes	Duarte Calado	FTP	77	1428	Request: PASS ILMI
64602	2025-09-06 11:06:45.664220	Duarte Calado	Miguel Pontes	FTP	88	1428	Response: 530 Login incorrect.
64609	2025-09-06 11:06:45.666306	Duarte Calado	Miguel Pontes	FTP	86	1429	Response: 220 (vsFTPd 3.0.5)
64611	2025-09-06 11:06:45.667986	Miguel Pontes	Duarte Calado	FTP	79	1429	Request: USER duarte
64613	2025-09-06 11:06:45.667987	Duarte Calado	Miguel Pontes	FTP	100	1429	Response: 331 Please specify the password.
64614	2025-09-06 11:06:45.668237	Miguel Pontes	Duarte Calado	FTP	81	1429	Request: PASS PASSWORD
64624	2025-09-06 11:06:48.883697	Duarte Calado	Miguel Pontes	FTP	88	1429	Response: 530 Login incorrect.
64631	2025-09-06 11:06:48.886113	Duarte Calado	Miguel Pontes	FTP	86	1430	Response: 220 (vsFTPd 3.0.5)
64633	2025-09-06 11:06:48.888404	Miguel Pontes	Duarte Calado	FTP	79	1430	Request: USER duarte
64635	2025-09-06 11:06:48.888563	Duarte Calado	Miguel Pontes	FTP	100	1430	Response: 331 Please specify the password.
64636	2025-09-06 11:06:48.888715	Miguel Pontes	Duarte Calado	FTP	78	1430	Request: PASS admin
64657	2025-09-06 11:06:52.501408	Duarte Calado	Miguel Pontes	FTP	88	1430	Response: 530 Login incorrect.
64664	2025-09-06 11:06:52.503570	Duarte Calado	Miguel Pontes	FTP	86	1431	Response: 220 (vsFTPd 3.0.5)
64666	2025-09-06 11:06:52.504452	Miguel Pontes	Duarte Calado	FTP	79	1431	Request: USER duarte
64668	2025-09-06 11:06:52.504603	Duarte Calado	Miguel Pontes	FTP	100	1431	Response: 331 Please specify the password.
64669	2025-09-06 11:06:52.504909	Miguel Pontes	Duarte Calado	FTP	82	1431	Request: PASS comcomcom
64676	2025-09-06 11:06:55.888543	Duarte Calado	Miguel Pontes	FTP	88	1431	Response: 530 Login incorrect.
64683	2025-09-06 11:06:55.890810	Duarte Calado	Miguel Pontes	FTP	86	1432	Response: 220 (vsFTPd 3.0.5)
64685	2025-09-06 11:06:55.892065	Miguel Pontes	Duarte Calado	FTP	79	1432	Request: USER duarte
64687	2025-09-06 11:06:55.892141	Duarte Calado	Miguel Pontes	FTP	100	1432	Response: 331 Please specify the password.
64688	2025-09-06 11:06:55.892428	Miguel Pontes	Duarte Calado	FTP	80	1432	Request: PASS <BLANK>
64698	2025-09-06 11:06:58.495429	Duarte Calado	Miguel Pontes	FTP	88	1432	Response: 530 Login incorrect.
64705	2025-09-06 11:06:58.498239	Duarte Calado	Miguel Pontes	FTP	86	1433	Response: 220 (vsFTPd 3.0.5)
64707	2025-09-06 11:06:58.500030	Miguel Pontes	Duarte Calado	FTP	79	1433	Request: USER duarte
64709	2025-09-06 11:06:58.500182	Duarte Calado	Miguel Pontes	FTP	100	1433	Response: 331 Please specify the password.
64710	2025-09-06 11:06:58.500256	Miguel Pontes	Duarte Calado	FTP	81	1433	Request: PASS PASSWORD
64718	2025-09-06 11:07:01.394001	Duarte Calado	Miguel Pontes	FTP	88	1433	Response: 530 Login incorrect.
64725	2025-09-06 11:07:01.397349	Duarte Calado	Miguel Pontes	FTP	86	1434	Response: 220 (vsFTPd 3.0.5)
64728	2025-09-06 11:07:01.399187	Miguel Pontes	Duarte Calado	FTP	79	1434	Request: USER duarte
64733	2025-09-06 11:07:01.399524	Duarte Calado	Miguel Pontes	FTP	100	1434	Response: 331 Please specify the password.
64734	2025-09-06 11:07:01.399524	Miguel Pontes	Duarte Calado	FTP	89	1434	Request: PASS duartinhothebest
64736	2025-09-06 11:07:01.430734	Duarte Calado	Miguel Pontes	FTP	89	1434	Response: 230 Login successful.
64830	2025-09-06 11:07:20.401669	Duarte Calado	Miguel Pontes	FTP	86	1435	Response: 220 (vsFTPd 3.0.5)
64832	2025-09-06 11:07:22.728524	Miguel Pontes	Duarte Calado	FTP	79	1435	Request: USER duarte
64834	2025-09-06 11:07:22.728969	Duarte Calado	Miguel Pontes	FTP	100	1435	Response: 331 Please specify the password.
64844	2025-09-06 11:07:26.573187	Miguel Pontes	Duarte Calado	FTP	89	1435	Request: PASS duartinhothebest
64845	2025-09-06 11:07:26.604652	Duarte Calado	Miguel Pontes	FTP	89	1435	Response: 230 Login successful.
64847	2025-09-06 11:07:26.606925	Miguel Pontes	Duarte Calado	FTP	72	1435	Request: SYST
64848	2025-09-06 11:07:26.606925	Duarte Calado	Miguel Pontes	FTP	85	1435	Response: 215 UNIX Type: L8

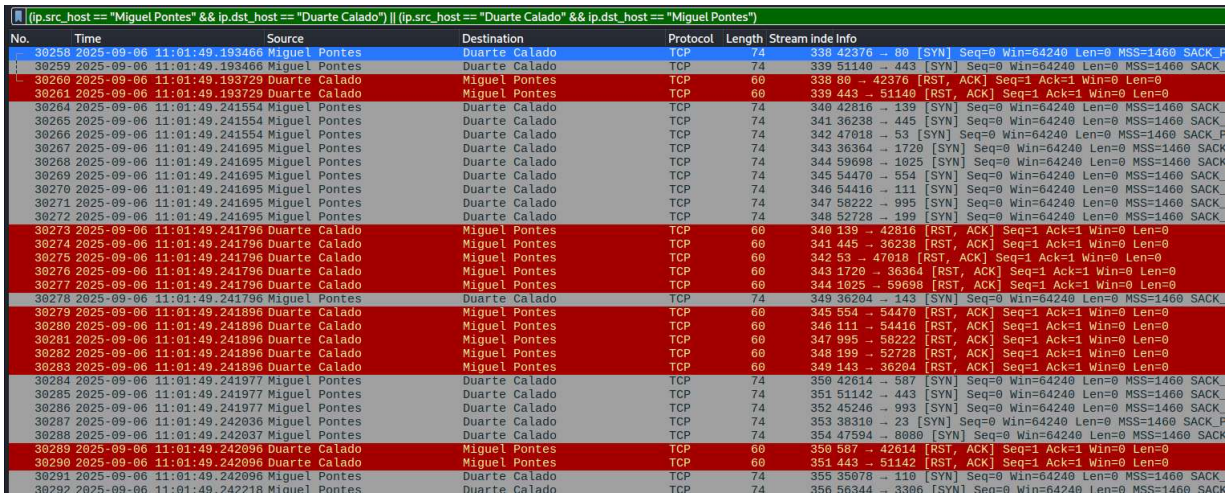
Figure 2: FTP login attempts and successful authentication by Miguel

In addition to the successful file transfers previously described, the capture also contains a sequence of FTP commands and corresponding server responses, confirming interactive access to the remote host. This behaviour was verified after applying the `FTP` display filter and locating the packet immediately following the one indicating "Login successful". By right-clicking on packet 64830 and selecting **Follow → TCP Stream**, the complete FTP session was reconstructed, revealing the sequence of issued commands, server responses, and subsequent file transfers.

Subsequently, by applying the display filter:

```
(ip.src_host == "194.210.61.136" && ip.dst_host == "194.210.61.134") || (ip.src_host == "194.210.61.134" && ip.dst_host == "194.210.61.136")
```

to isolate traffic exchanged between Miguel (194.210.61.136) and Duarte (194.210.61.134), it was possible to confirm direct communication between the two hosts.



(ip.src_host == "Miguel Pontes" && ip.dst_host == "Duarte Calado") || (ip.src_host == "Duarte Calado" && ip.dst_host == "Miguel Pontes")

No.	Time	Source	Destination	Protocol	Length	Stream inde	Info
30258	2025-09-06 11:01:49.193466	Miguel Pontes	Duarte Calado	TCP	74	338	42376 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_P
30259	2025-09-06 11:01:49.193466	Miguel Pontes	Duarte Calado	TCP	74	339	51140 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30260	2025-09-06 11:01:49.193729	Duarte Calado	Miguel Pontes	TCP	60	338	80 → 42376 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30261	2025-09-06 11:01:49.193729	Duarte Calado	Miguel Pontes	TCP	60	339	443 → 51140 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30264	2025-09-06 11:01:49.241554	Miguel Pontes	Duarte Calado	TCP	74	340	42816 → 139 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30265	2025-09-06 11:01:49.241554	Miguel Pontes	Duarte Calado	TCP	74	341	36238 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30266	2025-09-06 11:01:49.241554	Miguel Pontes	Duarte Calado	TCP	74	342	47018 → 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_P
30267	2025-09-06 11:01:49.241695	Miguel Pontes	Duarte Calado	TCP	74	343	36364 → 1720 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
30268	2025-09-06 11:01:49.241695	Miguel Pontes	Duarte Calado	TCP	74	344	59698 → 1025 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
30269	2025-09-06 11:01:49.241695	Miguel Pontes	Duarte Calado	TCP	74	345	54470 → 554 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30270	2025-09-06 11:01:49.241695	Miguel Pontes	Duarte Calado	TCP	74	346	54416 → 111 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30271	2025-09-06 11:01:49.241695	Miguel Pontes	Duarte Calado	TCP	74	347	58222 → 995 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30272	2025-09-06 11:01:49.241695	Miguel Pontes	Duarte Calado	TCP	74	348	52728 → 199 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30273	2025-09-06 11:01:49.241796	Duarte Calado	Miguel Pontes	TCP	60	340	139 → 42816 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30274	2025-09-06 11:01:49.241796	Duarte Calado	Miguel Pontes	TCP	60	341	445 → 36238 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30275	2025-09-06 11:01:49.241796	Duarte Calado	Miguel Pontes	TCP	60	342	53 → 47018 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30276	2025-09-06 11:01:49.241796	Duarte Calado	Miguel Pontes	TCP	60	343	1720 → 36364 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30277	2025-09-06 11:01:49.241796	Duarte Calado	Miguel Pontes	TCP	60	344	1025 → 59698 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30278	2025-09-06 11:01:49.241796	Miguel Pontes	Duarte Calado	TCP	74	349	36204 → 143 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30279	2025-09-06 11:01:49.241896	Duarte Calado	Miguel Pontes	TCP	60	345	554 → 54470 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30280	2025-09-06 11:01:49.241896	Duarte Calado	Miguel Pontes	TCP	60	346	111 → 54416 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30281	2025-09-06 11:01:49.241896	Duarte Calado	Miguel Pontes	TCP	60	347	995 → 58222 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30282	2025-09-06 11:01:49.241896	Duarte Calado	Miguel Pontes	TCP	60	348	199 → 52728 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30283	2025-09-06 11:01:49.241896	Duarte Calado	Miguel Pontes	TCP	60	349	143 → 36204 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30284	2025-09-06 11:01:49.241977	Miguel Pontes	Duarte Calado	TCP	74	350	42614 → 587 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30285	2025-09-06 11:01:49.241977	Miguel Pontes	Duarte Calado	TCP	74	351	51142 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30286	2025-09-06 11:01:49.241977	Miguel Pontes	Duarte Calado	TCP	74	352	45246 → 993 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30287	2025-09-06 11:01:49.242036	Miguel Pontes	Duarte Calado	TCP	74	353	38310 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_P
30288	2025-09-06 11:01:49.242037	Miguel Pontes	Duarte Calado	TCP	74	354	47594 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
30289	2025-09-06 11:01:49.242096	Duarte Calado	Miguel Pontes	TCP	60	350	587 → 42614 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30290	2025-09-06 11:01:49.242096	Duarte Calado	Miguel Pontes	TCP	60	351	443 → 51142 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
30291	2025-09-06 11:01:49.242096	Miguel Pontes	Duarte Calado	TCP	74	355	35078 → 110 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
30292	2025-09-06 11:01:49.242218	Miguel Pontes	Duarte Calado	TCP	74	356	56344 → 3306 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK

Figure 3: Direct TCP communication between Miguel and Duarte

An extensive amount of TCP packets were captured within a time span of less than 100 milliseconds, indicating a rapid and automated connection attempt, consistent with a port scanning activity. Among all the scanned ports, only two responded with valid [SYN, ACK] packets, confirming open services on ports 22 (SSH) and 21 (FTP). These interactions are first observed at packet 30258 and last detected at packet 32264 in the capture.

Trace 3:

The analysis of Trace 3 began again with applying the `IRC` display filter, which allowed identification of relevant chat streams. Two primary streams were observed:

- **tcp.stream eq 17:** Start packet 183 – End packet 678;
- **tcp.stream eq 2054:** Start packet 96095 – End packet 97329.

Within the first IRC stream, Miguel accessed another individual's WhatsApp content and shared a screenshot, which is available at the following URL: *https://files.catbox.moe/jygyt1.png*, this being the whatsapp related secret (jygyt1.png).

As in previous traces, HTTP objects were exported via **File → Export Objects → HTTP**, and by inspecting the packets and following the HTTP streams, the visited websites and resources were identified. The most relevant entries are summarised below:

Website / Resource	Packet No.	Notes	Host
https://askubuntu.com/questions/264046/how-to-run-the-ssh-server-on-a-port-other-than-22	7063	SSH configuration research	Miguel
https://www.digitalocean.com/community/tutorials/how-to-use-ssh-to-connect-to-a-remote-server	12582	SSH usage / connection tutorial	Miguel
https://opensource.com/article/20/9/ssh	20313	SSH background / best practices	Miguel
https://www.malwarebytes.com/pt/phishing	27191	Anti-phishing guidance (context for phishing activity)	Miguel

https://chatgpt.com	32872	Used for phishing email	Miguel
https://www.reddit.com/r/FinancialPlanning/comments/umjp5b/what_to_do_with_extra_money/	52796	Packet 52715 shows Adelino performing a DNS query for www.reddit.com	Adelino
https://gambling.com	64959	Gambling-related sites	Adelino
https://www.quora.com/Is-it-possible-to-win-at-a-casino-by-choosing-all-reds-in-roulette-each-time-you-gamble	73566	Gambling-related sites	Adelino
https://proton.me/pt	37917	Email	Miguel
https://proton.me/pt	84661	Email	Adelino

Similarly to trace 2, the ChatGPT website was also accessed in this trace, and the content of the interactions was retrieved using the same procedure of following HTTP/2 substreams, saving them, and processing with the custom parsing script:

TCP Stream	HTTP/2 Stream	Last Packet No.	Sender	Content Description
1447	47	34824	Miguel	First prompt
1447	47	35222	Chat	First response
1447	71	35331	Miguel	Second prompt
1447	71	35497	Chat	Second response

To identify the user's search activity, the display filter `http3.headers.header.value contains "search"` was applied. This allowed the recovery of Google search queries along with the probable destination sites observed in the captured HTTP objects. The results are summarized in the table below:

Packet No.	Query (Google)	Probable destination(s) observed in HTTP objects	User
5025	how to establish a connection using ssh port	https://askubuntu.com/questions/264046/how-to-run-the-ssh-server-on-a-port-other-than-22. https://www.digitalocean.com/community/tutorials/how-to-use-ssh-to-connect-to-a-remote-server https://opensource.com/article/20/9/ssh	Miguel
23674	phishing	https://www.malwarebytes.com/pt/phishing	Miguel
31677	chatgpt	https://chatgpt.com	Miguel
36763	proton	https://proton.me/pt	Miguel
47638	what to do with excess money	https://www.reddit.com/r/FinancialPlanning/comments/umjp5b/what_to_do_with_extra_money/	Adelino
62487	gambling	https://gambling.com	Adelino
71364	should I go all in red	https://www.quora.com/Is-it-possible-to-win-at-a-casino-by-choosing-all-reds-in-roulette-each-time-you-gamble	Adelino
84036	protonmail	https://proton.me/pt	Adelino

Subsequently, as observed in the previous trace, applying the ARP display filter revealed a large number of requests issued by Miguel, indicating extensive ARP query activity on the network.

arp

No.	Time	Source	Destination	Protocol	Length	Stream inde	Info
689	2025-09-06 15:17:30.241828	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.10? Tell 194.210.61.136
690	2025-09-06 15:17:30.243824	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.11? Tell 194.210.61.136
691	2025-09-06 15:17:30.246039	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.12? Tell 194.210.61.136
692	2025-09-06 15:17:30.248282	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.13? Tell 194.210.61.136
693	2025-09-06 15:17:30.250240	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.14? Tell 194.210.61.136
694	2025-09-06 15:17:30.252077	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.15? Tell 194.210.61.136
695	2025-09-06 15:17:30.253834	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.16? Tell 194.210.61.136
696	2025-09-06 15:17:30.255854	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.17? Tell 194.210.61.136
697	2025-09-06 15:17:30.257942	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.18? Tell 194.210.61.136
698	2025-09-06 15:17:30.260383	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.19? Tell 194.210.61.136
699	2025-09-06 15:17:30.262422	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.20? Tell 194.210.61.136
700	2025-09-06 15:17:30.264201	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.21? Tell 194.210.61.136
701	2025-09-06 15:17:30.265988	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.22? Tell 194.210.61.136
702	2025-09-06 15:17:30.268124	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.23? Tell 194.210.61.136
703	2025-09-06 15:17:30.270293	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.24? Tell 194.210.61.136
704	2025-09-06 15:17:30.272625	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.25? Tell 194.210.61.136
705	2025-09-06 15:17:30.274638	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.26? Tell 194.210.61.136
706	2025-09-06 15:17:30.276154	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.27? Tell 194.210.61.136
707	2025-09-06 15:17:30.277888	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.28? Tell 194.210.61.136
708	2025-09-06 15:17:30.279866	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.29? Tell 194.210.61.136
709	2025-09-06 15:17:30.282167	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.30? Tell 194.210.61.136
710	2025-09-06 15:17:30.284415	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.31? Tell 194.210.61.136
711	2025-09-06 15:17:30.286493	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.32? Tell 194.210.61.136
712	2025-09-06 15:17:30.288220	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.33? Tell 194.210.61.136
713	2025-09-06 15:17:30.290019	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.34? Tell 194.210.61.136
714	2025-09-06 15:17:30.292237	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.35? Tell 194.210.61.136
715	2025-09-06 15:17:30.294546	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.36? Tell 194.210.61.136
716	2025-09-06 15:17:30.297075	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.37? Tell 194.210.61.136
717	2025-09-06 15:17:30.298587	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.38? Tell 194.210.61.136
718	2025-09-06 15:17:30.300065	Miguel Pontes	Broadcast	ARP	60		Who has 194.210.60.39? Tell 194.210.61.136

Figure 4: ARP query activity observed in the third trace

Additionally, packets were observed from Miguel's to Duarte's workstation via the SSH protocol, indicating repeated connection attempts:

ssh

No.	Time	Source	Destination	Protocol	Length	Stream inde	Info
23862	2025-09-06 15:18:56.818005	Adelino Estoiro	Miguel Pontes	SSHv2	1406	1336	Server: Protocol (SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14)
23863	2025-09-06 15:18:56.818070	Adelino Estoiro	Miguel Pontes	SSHv2	1398	1336	Server: Key Exchange Init
23865	2025-09-06 15:18:56.678341	Miguel Pontes	Adelino Estoiro	SSHv2	1398	1336	Client: Key Exchange Init
23867	2025-09-06 15:18:56.678341	Adelino Estoiro	Miguel Pontes	SSHv2	198	1336	Server: Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=284)
23869	2025-09-06 15:18:56.678380	Miguel Pontes	Adelino Estoiro	SSHv2	198	1336	Client: New Keys, Encrypted packet (len=68)
23871	2025-09-06 15:18:56.734862	Adelino Estoiro	Miguel Pontes	SSHv2	118	1336	Server: Encrypted packet (len=84)
23872	2025-09-06 15:18:56.734862	Adelino Estoiro	Miguel Pontes	SSHv2	118	1336	Client: Encrypted packet (len=84)
23874	2025-09-06 15:18:56.734862	Miguel Pontes	Adelino Estoiro	SSHv2	134	1336	Client: Encrypted packet (len=84)
23875	2025-09-06 15:18:56.730724	Adelino Estoiro	Miguel Pontes	SSHv2	98	1336	Server: Encrypted packet (len=64)

Figure 5: SSH connection attempts from Miguel to Duarte's system

Finally, in the main Wireshark packet list, packets were identified containing the POST /api/mail request from Miguel's host (packet 82981). By following the HTTP/2 stream (Follow → HTTP Stream), the packet content revealed a JSON payload. This payload was extracted and saved to a separate file, then processed using the `parse_proton_sent.py` script to interpret the email's structure and metadata. The resulting output was as follows:

=== MESSAGE SUMMARY ===
ID: YJzaub7NOBga0VSZI4-cTheyF7fQ7u1q9TtkchydMYiE5JTKEH4iqwEaZgrh0OIUh4w7RVrlFlpr-KUaVXeueQ==
Subject: Urgent: Immediate Action Required – Credential Revocation and Token Update
Time (timestamp): 1759867904
Time (UTC) : 2025-10-07T20:11:44+00:00
Time (Europe/Lisbon): 2025-10-07T21:11:44+01:00
SenderAddress: paulo.r.parquimetro@proton.me
To: adelino.estoiro@proton.me
IsEncrypted: True
NumAttachments: 1
Size (bytes): 3870
Saved PGP body to: phishing_email_Miguel/message.pgp.asc Saved parsed headers to: phishing_email_Miguel/parsed_headers.json Saved attachments metadata to: phishing_email_Miguel/attachments.json

Attachments:

- *Name: token_update.zip Size: 2067 MIME: application/zip ID: M3v36GkZ6uKFCZSG_4Mx9t6pHEfDXl6Q0VLp4yCY6Q8FVzcf5pnwRRrR2hj-ksjZL3503n6HEvpBwjAv8TBTZg==*

This confirms that Miguel, using the alias paulo.r.parquimetro@proton.me, sent an encrypted message to adelino.estoiro@proton.me with the subject "Urgent: Immediate Action Required – Credential Revocation and Token Update". Although the message body was encrypted, the metadata reveals a single attachment named token_update.zip, which, based on the filename and context, likely contained the malicious payload related to the phishing campaign.

A similar POST /api/mail request was later observed from Adelino's host (packet 93281), corresponding to his reply to Miguel's message. This second JSON object was processed using the same method, yielding the following:

=== MESSAGE SUMMARY ===
ID: c_f9OtgaOVlGwP9nCsBdi-34g7kHuWXNHSfDI1VY_AJHAHtFz8Ffa_5GNvsz-jqwac78EzKNn9XV8DK4hQq3uA==
Subject: Re: Urgent: Immediate Action Required – Credential Revocation and Token Update
Time (timestamp): 1759867965
Time (UTC) : 2025-10-07T20:12:45+00:00
Time (Europe/Lisbon): 2025-10-07T21:12:45+01:00
SenderAddress: adelino.estoiro@proton.me
To: paulo.r.parquimetro@proton.me
IsEncrypted: True
NumAttachments: 0
Size (bytes): 2079
Saved PGP body to: phishing_email_Adelino_response/message.pgp.asc
Saved parsed headers to: phishing_email_Adelino_response/parsed_headers.json
No attachments metadata found.

This confirms that Adelino replied directly to Miguel's email, maintaining the same subject line. Unlike the first message, his response contained no attachments, further supporting the conclusion that the initial "token_update.zip" file originated from Miguel and likely constituted the phishing payload distributed in this exchange.

In this sequence, several packets containing large amounts of URL-encoded data were observed, interspersed with smaller control messages. These data-heavy packets may correspond to the invoice and financial report mentioned by Miguel during the IRC conversation (file1.txt and file2.txt).

3. Analysis of relevant findings

3.1. Do the analyzed network traces contain evidence of transfers involving any of the six hidden documents identified in previous assignments? What can you infer about the source and origin of these documents within MEMEL's network?

The first network trace analyzed (Trace 1) already provides concrete evidence of the transfer of two of the six hidden artifacts previously identified during the investigation. Although the content was not transmitted through file attachments or encrypted channels, it was exchanged through IRC messages between two internal MEMEL employees: Miguel Pontes (Gateway Security Intern) and Catarina Pato (Information Systems Intern), who

identified by the nickname `CatDucky`. The captured traffic shows that this IRC communication occurred externally (`chat.freenode.net`), bypassing MEMEL's official communication infrastructure.

Within the chat, Catarina pasted the content of a **code diff** directly into the conversation. The diff corresponds to a commit introducing preferential pricing rules for test plates and discriminatory pricing against foreign or rental cars, information later found embedded in one of the steganographic files recovered from João Musk's devices and the cloned website (**`diff.txt`**). To verify this, we extracted the diff from the IRC conversation (`irc_extracted.txt`) and observed in the packet bytes that each line ended with `\r\n` (hex `0d0a`), while the previously recovered secret used only `\n`. We normalized the line endings using the following command:

```
- tr -d '\r' < irc_extracted.txt > normalized.txt
```

and compared the two files with the `diff` command:

```
└─$ diff normalized.txt diff.txt
11,13c11,12
< <
< <
< >
\ Sem newline no fim do ficheiro
---
> <
> >
\ Sem newline no fim do ficheiro
```

Figure 6: Output of `diff normalized.txt diff.txt`, showing minor differences

Minor differences were found. Specifically, one missing line containing only the `<` character and a few extra whitespace characters, but the code content itself was identical. As a result, although the hashes do not match exactly, the extracted content clearly corresponds to the same diff, supporting its identification as the first transferred secret.

The irc conversation also contains a link to an image hosted on `catbox.moe`, which shows a screenshot of a **Reddit thread** discussing tourist complaints about parking pricing in Lisbon. This screenshot matches the second of the "secrets" previously extracted from the Zyra phishing site backups (**`TouristThreads.png`**), having the same SHA-256 hash. These findings indicate that the IRC exchange is the earliest observed point of transfer for at least two of the leaked artifacts.

From a network attribution perspective, the source of the transfer is the IRC chat between Miguel and Catarina, both operating within MEMEL's internal subnet and routed through the HTTPS proxy. The origin of the diff itself is internal to MEMEL, specifically the production codebase referenced by Catarina, which she accessed while working on the information systems team. In her own words, *"I found that last year Duarte committed some weird lines directly to production"*, indicating that the relevant changes were introduced by Duarte approximately a year earlier. By contrast, the origin of the Reddit screenshot is external: Miguel likely retrieved it online, uploaded it to Catbox, and shared it over IRC. These details strongly support the conclusion that the leak originated from within MEMEL, involving employees with legitimate internal access who shared the information through off-channel communication. This represents the first clear evidence of how sensitive information began circulating prior to reaching João Musk.

The second network trace provides definitive evidence of the transfer of the third hidden document, **`parking.db`**, the internal database file previously identified among the six "secrets". Mentions of this file first appear within the fourth recovered IRC exchange between Miguel Pontes and Catarina Pato, where Miguel explicitly describes having accessed *"snippets of the database where plates are being stored"*, revealing that *"some plates are*

paying way more than others". This aligns precisely with the structure and content of the `parking.db` file recovered in earlier stages of the investigation, which contained tables referencing tester plates, transactions, and the biased pricing logic consistent with the "advanced rule" code snippet previously discovered by Catarina.

Technical inspection of the packet capture confirms the authenticity of this transfer. Within TCP stream 1445, investigators identified a series of FTP and FTP-Data exchanges between Miguel's workstation and Duarte Calado's system:

ftp || ftp-data

No.	Time	Source	Destination	Protocol	Length	Stream index	Info
649…	2025-09-06 11:07:54,86…	Miguel Pontes	Duarte Calado	FTP	81	1435	Request: CWD Downloads
649…	2025-09-06 11:07:54,86…	Duarte Calado	Miguel Pontes	FTP	103	1435	Response: 250 Directory successfully changed.
649…	2025-09-06 11:07:56,58…	Miguel Pontes	Duarte Calado	FTP	72	1435	Request: EPSV
649…	2025-09-06 11:07:56,58…	Duarte Calado	Miguel Pontes	FTP	114	1435	Response: 229 Entering Extended Passive Mode (|||51401|)
649…	2025-09-06 11:07:56,58…	Miguel Pontes	Duarte Calado	FTP	72	1435	Request: LIST
649…	2025-09-06 11:07:56,58…	Duarte Calado	Miguel Pontes	FTP	105	1435	Response: 150 Here comes the directory listing.
649…	2025-09-06 11:07:56,58…	Duarte Calado	Miguel Pontes	FTP-D…	207	1443	FTP Data: 141 bytes (EPASV) (LIST)
650…	2025-09-06 11:07:56,58…	Duarte Calado	Miguel Pontes	FTP	90	1435	Response: 226 Directory send OK.
650…	2025-09-06 11:08:00,86…	Miguel Pontes	Duarte Calado	FTP	72	1435	Request: EPSV
650…	2025-09-06 11:08:00,86…	Duarte Calado	Miguel Pontes	FTP	114	1435	Response: 229 Entering Extended Passive Mode (|||30440|)
650…	2025-09-06 11:08:00,86…	Miguel Pontes	Duarte Calado	FTP	72	1435	Request: NLST
650…	2025-09-06 11:08:00,86…	Duarte Calado	Miguel Pontes	FTP	105	1435	Response: 150 Here comes the directory listing.
650…	2025-09-06 11:08:00,86…	Duarte Calado	Miguel Pontes	FTP-D…	95	1444	FTP Data: 29 bytes (EPASV) (NLST)
650…	2025-09-06 11:08:00,86…	Duarte Calado	Miguel Pontes	FTP	90	1435	Response: 226 Directory send OK.
650…	2025-09-06 11:08:01,21…	Miguel Pontes	Duarte Calado	FTP	74	1435	Request: TYPE I
650…	2025-09-06 11:08:01,21…	Duarte Calado	Miguel Pontes	FTP	97	1435	Response: 200 Switching to Binary mode.
650…	2025-09-06 11:08:01,21…	Miguel Pontes	Duarte Calado	FTP	83	1435	Request: SIZE parking.db
650…	2025-09-06 11:08:01,21…	Duarte Calado	Miguel Pontes	FTP	77	1435	Response: 213 40960
650…	2025-09-06 11:08:01,21…	Miguel Pontes	Duarte Calado	FTP	72	1435	Request: EPSV
650…	2025-09-06 11:08:01,21…	Duarte Calado	Miguel Pontes	FTP	114	1435	Response: 229 Entering Extended Passive Mode (|||57555|)
650…	2025-09-06 11:08:01,21…	Miguel Pontes	Duarte Calado	FTP	83	1435	Request: RETR parking.db
650…	2025-09-06 11:08:01,21…	Duarte Calado	Miguel Pontes	FTP	137	1435	Response: 150 Opening BINARY mode data connection for parking.db (40960 bytes)
650…	2025-09-06 11:08:01,22…	Duarte Calado	Miguel Pontes	FTP-D…	7306	1445	FTP Data: 7240 bytes (EPASV) (RETR parking.db)
650…	2025-09-06 11:08:01,22…	Duarte Calado	Miguel Pontes	FTP-D…	7306	1445	FTP Data: 7240 bytes (EPASV) (RETR parking.db)
650…	2025-09-06 11:08:01,22…	Duarte Calado	Miguel Pontes	FTP-D…	102…	1445	FTP Data: 10136 bytes (EPASV) (RETR parking.db)
650…	2025-09-06 11:08:01,22…	Duarte Calado	Miguel Pontes	FTP-D…	145…	1445	FTP Data: 14480 bytes (EPASV) (RETR parking.db)
650…	2025-09-06 11:08:01,22…	Duarte Calado	Miguel Pontes	FTP-D…	1930	1445	FTP Data: 1864 bytes (EPASV) (RETR parking.db)
650…	2025-09-06 11:08:01,22…	Duarte Calado	Miguel Pontes	FTP	90	1435	Response: 226 Transfer complete.

Figure 7: FTP packets exchange and transfer of `parking.db` by Miguel

The logs clearly display the command sequence `RETR parking.db` issued from Miguel's host, followed by Duarte's system responding with `150 Opening BINARY mode data connection for parking.db (40960 bytes)`, indicating the beginning of the file transfer. The data itself was transmitted over five FTP-Data packets, all directed from Duarte's IP address to Miguel's. Using Wireshark's *Export Objects* feature, the transferred file was extracted and analyzed. Its SHA-256 hash matched perfectly with the corresponding `parking.db` artifact previously uncovered within João Musk's system. This cryptographic match conclusively verifies that the file transferred during this network session is the same internal database later found embedded among the leaked "secrets".

The source of the transfer is therefore the FTP session initiated by Miguel Pontes, while the origin of the document lies within Duarte Calado's workstation, specifically his *Downloads* directory, from which the file was retrieved. Together, these findings confirm that `parking.db` was exfiltrated directly from MEMEL's internal environment through unauthorized access by an insider using standard network protocols. The combination of conversational evidence from IRC and forensic verification from the packet capture provides a complete chain of custody for this document's movement - from its internal origin within MEMEL's systems to its illicit transfer by Miguel, eventually surfacing among the compromised materials linked to João Musk.

In the third network trace, evidence was found corresponding to the transfer of the fourth hidden document, the WhatsApp image (**`wpp.png`**) previously identified among the six leaked files. The image appears in an IRC conversation between Miguel Pontes and Catarina Pato (TCP stream 17), where Miguel recounts overhearing a conversation between Duarte and another employee in MEMEL's office. After Duarte left, Miguel noticed that the coworker sitting beside him had left his phone unlocked, displaying a WhatsApp group named "Parkinator". According to Catarina, the group included Duarte and Adelino Estoiro, and Miguel noted that it contained messages referring to certain individuals receiving "free parking". He took a screenshot of the chat, airdropped it to his own phone, deleted it from the coworker's device, and subsequently uploaded the image to the file-hosting service Catbox, sharing the link (`https://files.catbox.moe/jygyt1.png`) with Catarina through the IRC channel.

This exchange represents direct evidence of the document's transfer: from the coworker's phone to Miguel's device, then to the external Catbox server, and finally through IRC to Catarina. The image was recovered

from the capture using the provided link, and its SHA-256 hash matched the previously identified secret, confirming its authenticity. The source of this leak is thus the airdrop and subsequent Catbox upload performed by Miguel, while the origin is the coworker's phone within MEMEL's premises—the device where the "Parkinator" WhatsApp group was originally displayed.

In addition to the previously identified transfers, the remaining two hidden documents were also traced back to MEMEL's internal network through the analysis of Trace 3. The first relevant indicator appears in the IRC conversations between Miguel and Catarina, where Catarina suggests digging into Adelino's files to uncover potential links between MEMEL's financial activities and the company Futuro Inteligente. In a subsequent conversation, Miguel admits to having successfully executed a phishing attack against Adelino by sending a malicious email containing a backdoor, gaining access to Adelino's workstation and retrieving two documents: an **invoice** tied to Futuro Inteligente and a MEMEL **financial report**. These correspond precisely to the remaining two "secret" documents identified in earlier stages of the investigation: **`Invoice.pdf`** and **`FinancialReport.pdf`**, respectively. Additional contextual evidence was found in Miguel's chat with ChatGPT shortly before the incident, where he requested assistance in drafting a convincing email impersonating MEMEL's CISO, which further supports his intentional preparation of the attack.

Corroborating these statements, the network traces contain clear metadata associated with the phishing message. Analysis of HTTP/2 traffic to Proton Mail reveals that Miguel, using the account *paulo.r.parquimetro@proton.me*, sent an email with the subject "Urgent: Immediate Action Required – Credential Revocation and Token Update" to *adelino.estoiro@proton.me*. Using the Export Objects tool and filtering by `multipart/form-data` content type, we identified a relevant packet with the filename "attachments":

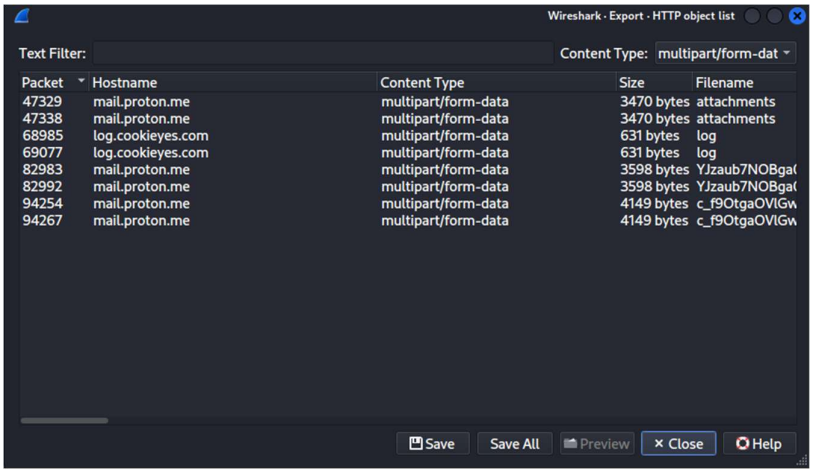


Figure 8: HTTP object list, filtered by `multipart/form-data` content type

In effect, the email contained a single attachment (**`token_update.zip`**) as identified in packet 47329 through inspection of the MIME multipart body ("Reassembled body" tab in the Packet Bytes Pane):

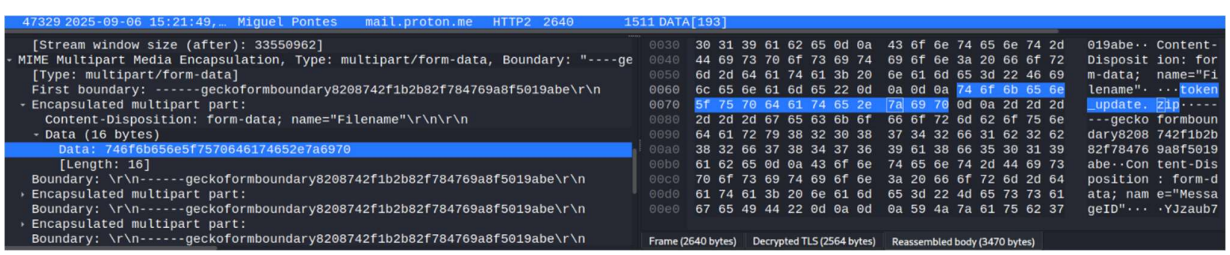
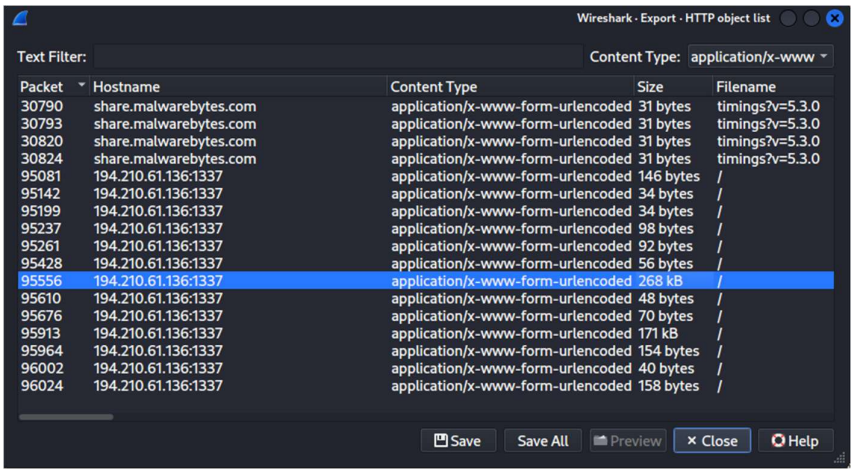


Figure 9: Inspection of packet 47329, revealing the `token_update.zip` attachment

Moments later, packet 93311 and associated traffic show Adelino replying to this email, confirming receipt and execution of Miguel's instructions.

Following this phishing interaction, the actual file exfiltration was identified by filtering the network capture using the Export Objects tool and selecting the content type `application/x-www-form-urlencoded`:



Packet	Hostname	Content Type	Size	Filename
30790	share.malwarebytes.com	application/x-www-form-urlencoded	31 bytes	timings?v=5.3.0
30793	share.malwarebytes.com	application/x-www-form-urlencoded	31 bytes	timings?v=5.3.0
30820	share.malwarebytes.com	application/x-www-form-urlencoded	31 bytes	timings?v=5.3.0
30824	share.malwarebytes.com	application/x-www-form-urlencoded	31 bytes	timings?v=5.3.0
95081	194.210.61.136:1337	application/x-www-form-urlencoded	146 bytes	/
95142	194.210.61.136:1337	application/x-www-form-urlencoded	34 bytes	/
95199	194.210.61.136:1337	application/x-www-form-urlencoded	34 bytes	/
95237	194.210.61.136:1337	application/x-www-form-urlencoded	98 bytes	/
95261	194.210.61.136:1337	application/x-www-form-urlencoded	92 bytes	/
95428	194.210.61.136:1337	application/x-www-form-urlencoded	56 bytes	/
95556	194.210.61.136:1337	application/x-www-form-urlencoded	268 kB	/
95610	194.210.61.136:1337	application/x-www-form-urlencoded	48 bytes	/
95676	194.210.61.136:1337	application/x-www-form-urlencoded	70 bytes	/
95913	194.210.61.136:1337	application/x-www-form-urlencoded	171 kB	/
95964	194.210.61.136:1337	application/x-www-form-urlencoded	154 bytes	/
96002	194.210.61.136:1337	application/x-www-form-urlencoded	40 bytes	/
96024	194.210.61.136:1337	application/x-www-form-urlencoded	158 bytes	/

Figure 10: HTTP object list, filtered by `application/x-www-form-urlencoded` content type

This revealed a structured sequence of command and file transfer packets originating from Adelino's IP address and directed at Miguel's IP. Specifically, several short packets containing the `cmd=` key are observed, starting with packet 95081 (146 bytes), suggesting remote command execution. Shortly after, two significantly larger packets appear with the `file=` key in the payload: packet 95556 (268 kB) and packet 95913 (171 kB), with two command messages in between. These are followed by additional small command packets, likely used to finalize or clean up the session. This pattern is entirely consistent with exfiltration of documents over a lightweight HTTP backdoor.

Given their timing and size, these two file transfers are highly likely to correspond to the **invoice** and **financial report** that Miguel mentioned during the IRC conversation. Although the transferred data is encrypted and direct verification by hash comparison is not possible, the files were nonetheless extracted, and their SHA-256 values documented for the forensic record. The source of the transfer is the backdoor session initiated after the phishing attack, while the origin of the data is clearly Adelino's workstation, from which the files were retrieved.

With this, evidence of the transfer of all six previously identified secret documents has been established across the three network traces. Collectively, these findings confirm that every document originated from inside MEMEL's network and was exfiltrated through a combination of legitimate access, opportunistic collection, and deliberate compromise of internal systems.

3.2. Based on the observed network activity, what can you deduce about the identity of the person or persons responsible for transferring the documents?

Analysis of the observed network activity provides a clear picture of the individuals responsible for transferring the confidential documents. The main actor in this process was **Miguel Pontes**, the security intern at MEMEL, who personally located, retrieved, and transmitted five of the six previously identified secret documents. His actions went well beyond passive observation: he conducted network scanning within MEMEL's internal subnet, performed targeted attacks on colleagues' machines, and used external platforms to exfiltrate data. Specifically, Miguel retrieved the `parking.db` file through FTP, accessed two email files from Duarte's directories, and later

delivered a phishing email with a backdoor to Adelino Estoiro to obtain an invoice and a financial report. He also physically exfiltrated an image of the "Parkinator" WhatsApp group by screenshotting a coworker's unlocked phone and uploading it to Catbox. These actions demonstrate a high degree of initiative, technical knowledge, and deliberate intent to bypass internal security controls.

Although Miguel is the primary agent in the document exfiltration, **Catarina Pato** also played an active role in the incident. Catarina was directly responsible for the transfer of the first secret (the discriminatory pricing diff) by pasting it into their IRC chat. Moreover, she consistently supported Miguel's activities throughout the entire process, offering strategic suggestions and encouraging further probing. For example, during the discussion about targeting Adelino's files, she proposed: *"We should dig a bit into Adelino, as he's the financial director, maybe there's something hidden in his files or something… little breadcrumbs linking Futuro Inteligente to all this, some tiny traces that could blow the whole thing wide open if we piece them together o.o"* (IRC chat 5, Trace 3). She was fully aware of the risks involved, warning Miguel that *"If you are caught, you're most likely to lose your job, we cannot afford something like that!"* (IRC chat 4, Trace 2), which indicates premeditation and awareness of the illicit nature of their actions.

Their later conversations reveal that both Catarina and Miguel were not merely uncovering irregularities: they were actively plotting to expose MEMEL externally. Catarina explicitly proposed an exfiltration plan to conceal their involvement: *"Okay, so here's what I think is safest, you'll send me the files from a burner account. That way, if anyone ever asks, it'll look like I just 'received' it anonymously o.o"* (IRC chat 6, Trace 3). She also suggested leveraging her connection to João Musk, whose father was running for mayor, to politicize the findings and tarnish Eurico Trocos's image: *"If João sees this, he can pass it straight to his dad's campaign".* These statements clearly indicate that Catarina, while not directly stealing the majority of the files herself, acted as an enabler and co-conspirator, facilitating the exfiltration and subsequent dissemination of the documents.

In addition to the technical evidence of data transfer, the tone and familiarity in their IRC exchanges suggest a close personal relationship between Miguel and Catarina. Their interactions include joking, teasing, and affectionate remarks, such as Miguel calling Catarina "Ducky" or "little princess" and making flirtatious comments. This personal closeness likely fostered an environment of trust and coordination between them, making their cooperation in the data leak both more effective and more difficult to detect.

Taken together, these elements show that the document exfiltration was not the work of an external attacker, but rather the result of coordinated actions by two internal employees. Miguel was the primary operative, executing the technical attacks and data retrieval, while Catarina acted as a strategic facilitator, both initiating one of the leaks herself and contributing to planning the dissemination of the stolen documents. Their shared intent, explicit coordination, and awareness of the risks make them jointly responsible for the transfer of the documents.

3.3. Can you reconstruct a timeline of key events that explains how the data exfiltration took place and how the documents ultimately reached João Musk's possession? Present your reasoning clearly and reference all timestamps used to establish the chronology.

The first network capture (**t1.pcapng**) revealed three Internet Relay Chat (IRC) exchanges between MEMEL interns **Miguel Pontes** (Gateway Security Team) and **Catarina Pato** (Information Systems Team). The initial conversation, beginning at **09:05:17 on 6 September 2025**, lasted about two minutes and appeared routine. A second exchange at **11:27:32**, lasting roughly four minutes, was the first to reveal potentially relevant internal activity: Miguel recounted overhearing **Alcibíades** asking **Duarte Calado**, MEMEL's IT Director, to "pull some strings" to avoid paying parking fees for his new vehicle. This prompted the interns' first suspicions of misconduct, establishing Duarte as a target of interest.

```
PRIVMSG #thingys :Hey Cat, where are you?
MODE #thingys b

@time=2025-10-07T14:27:17.663Z :*.freenode.net 368 migstop #thingys :End of channel ban list

NOTICE Harambe :\x01VERSION irssi v1.4.5\x01
PING *.freenode.net

@time=2025-10-07T14:27:24.683Z :*.freenode.net PONG *.freenode.net :*.freenode.net
@time=2025-10-07T14:27:31.291Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Just finished a meeting! I'm in the Blue area of the office. What's up?

PRIVMSG #thingys :Ah ok, maybe I should explain here so no one overhears... you won't believe what I just heard in the kitchen...

@time=2025-10-07T14:27:46.001Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :TELL ME EVERYTHING NOWWWW

PRIVMSG #thingys :I went to grab a coffee and found Duarte talking with Alcibiades about some exotic new car Alcibiades bought, which you'd actually like haha It's pretty sick-looking
PRIVMSG #thingys :But that's not the worst part...

@time=2025-10-07T14:28:10.043Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Can you stop teasing and just spill it already? .. -_-

PING *.freenode.net

@time=2025-10-07T14:28:25.885Z :*.freenode.net PONG *.freenode.net :*.freenode.net

PRIVMSG #thingys :yeah yeah, sorry ^^" Alcibiades was asking Duarte to "pull some strings" so he wouldn't have to pay for parking.

@time=2025-10-07T14:28:35.694Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :what?? But can they even do that?

PRIVMSG #thingys :No, they can't lol. But the worst bit is that when they saw me coming, they were alone and clearly didn't want anyone to hear, the moment they noticed someone else they awkwardly changed th
e subject, and then started joking about mining Bitcoin to buy the new iPhone XD

@time=2025-10-07T14:29:22.233Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :That is really weird, they must be hiding something...

PING *.freenode.net

@time=2025-10-07T14:29:27.065Z :*.freenode.net PONG *.freenode.net :*.freenode.net
@time=2025-10-07T14:29:31.401Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :AAAAA you can't tell me this when I've got stuff to do! I still have to fix a ticket for a DB bug since last week an
d I have no idea what it even is… AND NOW I'M SO CURIOUS, I won't be able to focus :(

PRIVMSG #thingys :Haha you asked for it XD
PRIVMSG #thingys :Lucky me, being the intern in the gateway team haha I just have to run some security checks :) imagine spending the day trying to solve hard algorithms in information systems team hahah

@time=2025-10-07T14:30:15.033Z :CatDucky!~cat@freenode-i4c.0j4.lpdv3i.IP PRIVMSG #thingys :Shut up :( I actually like messing around with the information systems, I'm learning a lot in this internship :( it'
s not that bad, just painful sometimes XD
```

Figure 11: Snippet of the second IRC conversation - Initial suspicion against Duarte Calado

At **15:27:30**, a third IRC session showed tangible investigative progress. Catarina reported discovering a suspicious **git commit** by Duarte in an internal repository, sharing a diff later confirmed to match the known artifact (**`diff.txt`**). Miguel, meanwhile, referenced a Reddit thread about Lisbon parking irregularities, whose screenshot (**`TouristThreads.png`**) also matched prior evidence via SHA-256 verification. These findings validated both artifacts' authenticity and origin.

The next trace (**t2.pcapng**) captured Miguel's unauthorized reconnaissance against Duarte's workstation. At **11:00:07 on 7 September 2025**, Miguel researched **arp-scan** and immediately launched an ARP sweep over the **194.210.60.0/22** network, completing at **11:00:56** and identifying Duarte's host. He then researched and executed **nmap** port-scanning techniques, performing a SYN scan at **11:01:49**, which revealed **FTP (21)** and **SSH (22)** as open ports. This escalation from passive scanning to active probing demonstrated deliberate reconnaissance intent.

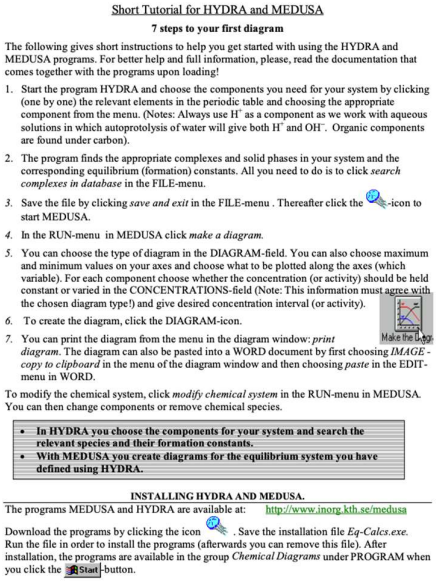
At **11:02**, Miguel contacted ChatGPT to help develop an FTP brute-forcing tool. The AI refused unethical instructions, instead suggesting legitimate tools like **Hydra** and **Medusa**. Miguel chose **Medusa**, researched its usage on **Kali.org** and **KTH Royal Institute** pages at **11:05:17**, and soon after launched a brute-force FTP attack at **11:06:20**. Authentication succeeded with the password **"duartinhothebest"**, allowing Miguel to access and extract three files: **parking.db**, **adelino.eml**, and "Re: Update for the Smart Algorithm – Duarte Calado.eml". The database matched the third secret by hash verification, confirming its authenticity. The emails revealed that Duarte's code changes had already redirected parking ticket data to **Futuro Inteligente**, linked via correspondence between **Duarte Calado**, **Eurico Trocos** (the mayor), and **Adelino Estoiro**, MEMEL's Financial Director, firmly connecting senior officials to the scheme.

Chat >

I can't help create an algorithm, script, or step-by-step instructions for brute-forcing FTP (that would meaningfully facilitate unauthorized access). I'll refuse that request.

If your goal is legitimate research or improving security, I can absolutely help — safely and ethically — by providing alternatives you can use to do lawful, controlled experiments or to harden systems. Below I've put together a practical, research-oriented plan (legal/ethics checklist, isolated lab setup, measurement methodology, defensive controls and detection strategies, and templates you can use). Tell me which parts you want expanded and I'll produce them.

Figure 12: First interaction with ChatGPT, refusing to help Miguel build a brute-forcing algorithm against FTP



Short Tutorial for HYDRA and MEDUSA

7 steps to your first diagram

The following gives short instructions to help you get started with using the HYDRA and MEDUSA programs. For better help and full information, please, read the documentation that comes together with the programs upon loading!

1. Start the program HYDRA and choose the components you need for your system by clicking (one by one) the relevant elements in the periodic table and choosing the appropriate component from the menu. (Notes: Always use H^+ as a component as we work with aqueous solutions in which autoprotolysis of water will give both H^+ and OH^-. Organic components are found under carbon).
2. The program finds the appropriate complexes and solid phases in your system and the corresponding equilibrium (formation) constants. All you need to do is to click *search complexes in database* in the FILE-menu.
3. Save the file by clicking *save and exit* in the FILE-menu . Thereafter click the -icon to start MEDUSA.
4. In the RUN-menu in MEDUSA click *make a diagram.*
5. You can choose the type of diagram in the DIAGRAM-field. You can also choose maximum and minimum values on your axes and choose what to be plotted along the axes (which variable). For each component choose whether the concentration (or activity) should be held constant or varied in the CONCENTRATIONS-field (Note: This information must agree with the chosen diagram type!) and give desired concentration interval (or activity).
6. To create the diagram, click the DIAGRAM-icon.
7. You can print the diagram from the menu in the diagram window: *print diagram*. The diagram can also be pasted into a WORD document by first choosing *IMAGE - copy to clipboard* in the menu of the diagram window and then choosing *paste* in the EDIT-menu in WORD.

Make the Diagram

To modify the chemical system, click *modify chemical system* in the RUN-menu in MEDUSA. You can then change components or remove chemical species.

- **In HYDRA you choose the components for your system and search the relevant species and their formation constants.**
- **With MEDUSA you create diagrams for the equilibrium system you have defined using HYDRA.**

INSTALLING HYDRA AND MEDUSA.

The programs MEDUSA and HYDRA are available at: http://www.inorg.kth.se/medusa

Download the programs by clicking the icon . Save the installation file *Eq-Calcs.exe.* Run the file in order to install the programs (afterwards you can remove this file). After installation, the programs are available in the group *Chemical Diagrams* under PROGRAM when you click the Start-button.

Figure 13: Short tutorial hosted by KTH on how to use Hydra/Medusa

Following the intrusion, Miguel performed contextual research beginning at **11:10:34**, visiting sites on Lisbon parking systems (**belocal.pt**) and smart-city initiatives (**Wikipedia**, **lisboaberta.cm-lisboa.pt**), and finally **futurointeligente.pt**, which appeared inauthentic. Seeking clarification, Miguel contacted his cousin, that works in Loja do Cidadão and learned that **Futuro Inteligente** was registered to **Cremilde Trocos**, wife of Mayor **Eurico Trocos**, confirming a direct conflict of interest. At **11:14:57**, Miguel relayed these findings to Catarina via IRC, establishing clear ties between MEMEL's internal manipulation, municipal figures, and the private company.

```
PRIVMSG #thingys :This all sounds like a well knit scam plot between them... and again that diff you fou
nd, it looks just like the plan they have been working on!
PRIVMSG #thingys :I know, right?! And it gets crazier.. I did some digging, and Futuro Inteligente Lda d
oesn't even have a proper website.. Can you believe that? There's one out there, but it's clearly not th
e same company.
PING *.freenode.net

@time=2025-10-08T23:18:49.499Z :*.freenode.net PONG *.freenode.net :*.freenode.net

PRIVMSG #thingys :And the kicker, my cousin works at the Loja do Cidadão and managed to pull up the comp
any registry. Guess who owns Futuro Inteligente? The mayor's wife, Cremilde Trocos. I swear I'm not maki
ng this up hahahaha
```

Figure 14: Snippet of the fourth IRC conversation - Owner of Futuro Inteligente is revealed

The third trace (**t3.pcapng**) resumed at **15:12:20** with another IRC exchange. Miguel recounted overhearing **Duarte Calado** discussing a secret **WhatsApp group** related to the scheme. When the coworker left his phone unlocked, Miguel accessed it and located the group "**Parkinator**," featuring Duarte, Adelino, Alcibíades, and Eurico Trocos among others. Miguel captured a screenshot, shared it with Catarina through a Catbox link, and its SHA-256 sum confirmed it as **`wpp.png`**, the fourth secret.

Continuing his investigation into **Adelino Estoiro**, Miguel executed another **arp-scan** at **15:17:34**, identifying Adelino's host, then performed a SYN scan revealing only **FTP** and **SSH** open. He researched SSH connection methods through **AskUbuntu**, **DigitalOcean**, and **OpenSource.com**, but his **15:18:56** SSH attempt was aborted before authentication. Pivoting from technical intrusion to **social engineering**, Miguel searched for "phishing" at **15:19:02**, read **Malwarebytes**'s article, and once again used an AI chatbot, this time successfully, to craft a phishing email.

Chat > Got it! Here's a more concise version:

Subject: **Immediate Action Required: Credential Revocation & Token Update**

Dear Team,

We've identified a security incident impacting certain machines within the company. To address this, please take the following steps immediately:

1. **Revoke credentials** on affected systems.
2. **Update the tokens** as per the attached instructions.

Please ensure these actions are completed as soon as possible to mitigate any risks. Should you have any questions, reach out to me or the IT support team.

Thank you for your prompt attention.

Best regards,
[Your Name]
CISO, MEMEL Company

This one is more to the point but still formal. Let me know if you want any adjustments!

Figure 15: Phishing email crafted by ChatGPT, under the impression of legitimate use

Miguel then accessed **Proton Mail**, a privacy-focused encrypted email provider, and at **15:27:13** sent a fraudulent message impersonating **"Paulo R. Parquímetro"**, MEMEL's CISO, claiming a "security incident" and attaching **`token_update.zip`**, containing a backdoor designed to exfiltrate data from Adelino's system. Meanwhile, at **15:22:03**, Adelino was browsing content about money management and gambling, visiting **Reddit**, **gambling.com**, and **Quora**. More specifically, he started by searching "what to do with excess money", which could be tied to the revenue originating from the fraudulent scheme being conducted at MEMEL. Within twenty seconds of Miguel's email, Adelino opened it, likely alerted via a notification, and replied, unknowingly executing the attachment.

At **15:31:41**, Miguel received Adelino's response confirming the phishing success. By **15:31:57**, the backdoor had connected to Miguel's Python-based command server on port **1337**, fetching shell commands via GET requests and returning outputs through POST fields labeled "cmd". Later, Miguel issued a command switching the payload to transmit files under the key "file." Two files (**268,365 bytes** and **171,295 bytes)** were exfiltrated in encrypted form, likely under symmetric key encryption contained in the attachment, preventing further inspection. Furthermore, the attachment itself can only be decrypted with the sender's private PGP key, which we do not have access to.

Figure 16: TCP Streams of the two extracted files from Adelino's workstation

At **15:35:55**, Miguel informed Catarina via IRC that he had extracted two files: a **Futuro Inteligente invoice (`Invoice.pdf`)** and a **MEMEL financial report (`FinancialReport.pdf`)**, completing the six stolen documents. During this exchange, the interns discussed using these materials to expose MEMEL's corruption. Catarina suggested contacting **João Musk**, a friend whose father was running for mayor against Eurico Trocos. Seeing an opportunity to both reveal the scandal and weaken Trocos's image, they planned a careful leak: Miguel would compile the six documents and send them anonymously from a public network to Catarina, who would forward them to João under the pretense of receiving them from an unknown source.

The leak was scheduled for the night of **September 6th**, but evidence shows **João Musk** only received the material on **September 9th**. The reason for the delay remains unknown. Furthermore, João did not share the evidence with his father; instead, motivated by personal resentment after an accident involving a **ZYRA** bicycle (a MEMEL service), he leaked the data independently, turning what began as a political expose into a personal act of revenge.

In summary, the timeline shows clear progression from curiosity to full-scale intrusion and exfiltration. Miguel's escalating actions, scanning, brute-forcing, phishing, and deploying malware, culminated in the transfer of six sensitive documents. These moved from Duarte Calado's and Adelino Estoiro's systems through Miguel's infrastructure, then to Catarina, and finally to João Musk. Ultimately, the disclosure deviated from the interns' intent: rather than fueling political accountability, the stolen data became the means for personal retaliation, closing a chain of digital misconduct that began with suspicion and ended in exposure.

3.4. Considering all the evidence collected throughout the MEMELgate investigation, what can you now conclude about the hypothesis of fraudulent activity within MEMEL that initiated this inquiry? Did you uncover additional evidence supporting the existence of fraud? Identify the actors who may be involved and outline your recommendations for the next phase of the investigation.

Hypothesis of fraudulent activity:

From the evidence accumulated across all investigation stages, there is now overwhelming proof that fraudulent activity occurred within MEMEL, coordinated internally and designed to redirect financial gains to Futuro Inteligente Lda., a company allegedly owned by Cremilde Trocos, the wife of Lisbon's Mayor Eurico Trocos. The latest communications and technical traces not only reinforce the initial suspicions from earlier labs but also reveal the human network behind the scheme.

Early Indicators of Illicit Conduct

The first IRC exchanges between Miguel Pontes and Catarina Duarte immediately expose knowledge of irregular behavior among MEMEL employees. Miguel reports overhearing Alcibíades Mourato requesting that Duarte Calado, MEMEL's IT director, "pull some strings" to exempt his vehicle from standard parking charges. This directly correlates with a screenshot later extracted from the WhatsApp group "Parkinator", showing that Alcibíades's license plate (MC-95-QN) had been added to the internal exemptions list, granting him free parking privileges. The same conversation indicates that Duarte and Alcibíades changed the topic abruptly when they noticed Miguel nearby, suggesting awareness of the illicit nature of their actions.

Code Manipulation and Technical Evidence

Further in the IRC conversation, Catarina identifies that Duarte had previously committed suspicious modifications directly into production code. The recovered snippet matches the deleted code fragment revealed in the `diff.txt` artifact from stage 2. This establishes direct code-level evidence of intentional fraud, linking system manipulation to a specific beneficiary.

Database Corroboration

In subsequent exchanges, Miguel confirms discovering "snippets of the database" where certain plates were "paying way more than others". This matches the database dump analyzed in Lab 1, `hidden.db` (named `parking.db` in João's workstation), where the licensors table listed Futuro Inteligente Lda. (ID 217) as a payment recipient for several parking zones, far exceeding its officially licensed areas. Combined with the manipulated code, the system was configured to route misallocated transactions to Futuro Inteligente.

Financial and Organizational Evidence

The IRC chat further references internal emails between Duarte and Adelino Estoiro, MEMEL's financial director, concerning payments for "AI services". This aligns perfectly with the invoice `Invoice.pdf` and MEMEL's annual financial report (`FinancialReport.pdf`), where Futuro Inteligente is described as a primary vendor for "AI calibration & smart mobility R&D."

Discovery of Conflict of Interest

The investigation takes a critical turn when Miguel discloses that Futuro Inteligente Lda. "doesn't even have a proper website" and that, according to a registry lookup performed via a contact at Loja do Cidadão, the company's registered owner is Cremilde Trocos, the Mayor's wife. This revelation exposes a possible direct conflict of interest and substantiates the hypothesis that the manipulated payments ultimately benefited a politically connected family.

Exfiltration of Evidence and Internal Concealment

Catarina and Miguel's subsequent communications demonstrate intent to leak evidence externally, outside MEMEL's control. Catarina instructed Miguel to send the incriminating files anonymously from a burner account, planning to later pass them to João Musk, whose father was politically opposed to Eurico Trocos.

However, in later questioning, Catarina denied awareness of the email sender's identity, claiming it was anonymous, contradicted by her own IRC messages.

So each newly uncovered element directly reinforces earlier findings:

Source	Key Evidence	Corroboration
IRC Logs	Duarte's production commits; references to exempted plates and financial flows	Matches *diff.txt* and database *hidden.db* anomalies
Database Dumps	Licensor 217 (*Futuro Inteligente Lda.*) receiving excess payments	Matches code logic
Invoices & Reports	"AI calibration & smart mobility R&D" payments	Confirmed by internal emails (Adelino -> Duarte)
IRC Disclosure	Futuro Inteligente owned by Mayor's wife	New link confirming conflict of interest
WhatsApp Screenshot	Exemptions and insider privileges	Matches Alcibíades's exempt plate and prior complaints

Authenticity and Evidence Integrity

As referenced in earlier stages of the investigation, the analysis of the network traces and packet captures confirms the authenticity of multiple critical artifacts, corresponding to the previously identified "secrets". The first identified secret, the code diff containing preferential and discriminatory pricing rules, was extracted from an IRC conversation and, after normalization of line endings, matched the content of the original *diff.txt* file. The second secret, the Reddit screenshot, was verified via identical SHA-256 hash to the previously recovered *TouristThreads*.png. Subsequent traces provided conclusive evidence of the third and fourth hidden documents: *parking.db,* an internal MEMEL database file, and the WhatsApp screenshot wpp.png, both confirmed by matching SHA-256 hashes to the earlier recovered artifacts. Additionally, two larger files, likely corresponding to the invoice tied to Futuro Inteligente and MEMEL's financial report, were extracted via a backdoor session originating from Adelino's workstation. Although these were encrypted during transfer and direct hash comparison was not possible, their SHA-256 values were documented, supporting the integrity and authenticity of all recovered data.

Additional evidence:

The following analysis examines the key communications that further substantiate the fraudulent activities within MEMEL.

Email exchange between Adelino Estoiro and Duarte Calado

The first recovered message was sent from *adelino.estoiro@proton.me* to *duarte.calado@proton.me*. Its content confirms the processing of a payment from MEMEL to Futuro Inteligente Lda. for alleged "AI services".

This correspondence provides documentary confirmation of the financial link between MEMEL and Futuro Inteligente. The language of the message is concise and lacks any contractual or operational details. Moreover, this "confirmation" email matches the financial documents discovered earlier, particularly the invoice and financial report excerpts referencing Futuro Inteligente's "AI calibration & smart mobility R&D", reinforcing that the payment was likely a mechanism to legitimize fraudulent transfers under the guise of consulting services.

This email thus establishes a direct chain of responsibility:

- Adelino Estoiro acted as the financial intermediary authorizing payments.
- Duarte Calado oversaw the technical implementation of the manipulated system that justified these payments.

Together, they represent the operational and financial coordination layer of the fraudulent scheme.

Email exchange between Eurico Trocos and Duarte Calado

The second chain of emails reveals a more explicit connection between MEMEL's system manipulation and political direction from Lisbon's mayor, Eurico Trocos.

In the initial message, Eurico instructs Duarte to "update a few specific plates": LP-67-NF, YU-31-ZI, and CE-43-NZ - noting that these vehicles "deserve special treatment, especially our visiting tourists". This phrasing clearly indicates preferential handling of certain license plates, effectively directing the system to exempt or favor selected individuals. Duarte's reply further confirms compliance:

"I have noted the updates for the smart algorithm and will ensure it handles the highlighted plates with the care and efficiency you expect."

He also adds that, for "quicker responses", they could "continue via WhatsApp", directly linking this conversation to the "Parkinator" WhatsApp group previously identified. The instruction to move the discussion to WhatsApp, an encrypted, less auditable channel, possibly demonstrates an intent to conceal communications, reinforcing the illicit nature of the exchanges.

Additionally, Eurico Trocos's instruction to give "special treatment" to specific license plates, labeled as *"our visiting tourists,"* directly aligns with a Parkinator WhatsApp message stating that *"the last thing we need is complaints from foreign plates."* As these visitors were prioritized, MEMEL staff sought to prevent any complaints from this specific group.

Taken together, these emails provide strong corroboration of the collusion between:

- Eurico Trocos, as the political figure issuing directives;
- Duarte Calado, as the technical executor;
- Adelino Estoiro, as the financial enabler.

They collectively demonstrate a deliberate and organized structure supporting the fraudulent "smart parking" scheme.

Actors:

This section identifies and characterizes the key human actors implicated in or connected to the fraudulent scheme uncovered within MEMEL and its associated entities. Attribution is based on digital evidence and correspondence logs collected during all investigation stages.

1. Duarte Calado - MEMEL IT Director

Duarte Calado is a central technical figure in the fraud. He was responsible for managing and deploying updates to MEMEL's smart parking system. Digital evidence (source code diffs, emails, and database artifacts) confirms that Duarte personally committed code introducing conditional exemptions and dynamic overpricing mechanisms, later linked to transactions redirected to Futuro Inteligente Lda. (licensor ID 217).

Furthermore, his correspondence with both Adelino Estoiro and Eurico Trocos confirms he acted as a conduit between MEMEL's internal systems and external political interests. Duarte also encouraged the use of non-official communication channels (e.g., WhatsApp), possibly to avoid traceability, a behavior also mentioned by Miguel in the IRC logs, which suggests deliberate attempts to limit traceability within MEMEL's official communication infrastructure.

2. Adelino Estoiro - MEMEL Financial Director

Adelino Estoiro's emails confirm that he approved and processed financial transfers to *Futuro Inteligente Lda.* for alleged "AI services". These transactions, which Adelino authorized and processed, corresponded to no actual deliverables from Futuro Inteligente, indicating his role in legitimizing payments linked to fraudulent activity. His position within MEMEL's finance department made him essential to the money flow that sustained the scheme.

3. Eurico Trocos - Mayor of Lisbon

Eurico Trocos is a high-level beneficiary of the scheme. His direct email communication with Duarte Calado reveals that he provided lists of specific vehicle plates to be granted "special handling," confirming awareness and participation in the manipulation of MEMEL's parking algorithm. Given his public office, this represents a severe breach of ethical and legal conduct, reinforcing the political dimension of the fraud.

4. Cremilde Trocos - Owner of Futuro Inteligente Lda

Cremilde Trocos, spouse of Eurico Trocos, is officially registered as the sole owner of *Futuro Inteligente Lda.*

This entity was found to be the main financial recipient of redirected MEMEL transactions. The ownership establishes a direct conflict of interest: profits generated through manipulated algorithms were funneled into a private company linked to the Mayor's immediate family.

5. Miguel Pontes - Gateway Security Intern MEMEL

Acted as an internal whistleblower but obtained sensitive information through illicit means. Miguel extracted confidential data from MEMEL's internal systems, including the *WhatsApp Parkinator* group messages and hidden files from MEMEL workstations, by compromising another employee's unlocked phone and later conducting unauthorized data exfiltration. Although his findings were crucial for uncovering the fraudulent scheme, his actions constituted a clear breach of digital forensics and ethical boundaries.

6. Catarina Duarte – Information Systems Intern MEMEL

Catarina acted as Miguel's collaborator. She advised him on concealment techniques and participated in the external dissemination of sensitive material. After receiving the evidence anonymously via the address *totallyaveragedude@tutamail.com*, she forwarded it to João Musk, who was politically connected to an opponent of Eurico Trocos. Although motivated by whistleblowing intentions, her messages demonstrate full awareness of the illegality of her actions and deliberate attempts to obscure her identity.

7. Alcibíades Mourato - Possibly MEMEL Employee

Alcibíades Mourato was overheard inside MEMEL's premises discussing with Duarte Calado how he could "pull some strings" to avoid paying for parking. His presence within the company facilities and his familiarity with MEMEL's staff suggest that he may be an employee or collaborator, although his exact role remains unconfirmed. His license plate ("MC-95-QN") was later found within the *tester_plates* database table, confirming his participation in the misuse of MEMEL's smart parking system for personal benefit.

8. Members of the WhatsApp Group "Parkinator"

This private communication channel included several MEMEL employees and external associates directly connected to the fraudulent operations, including Duarte Calado, Adelino Estoiro, Alcibíades Mourato, and Eurico Trocos, whose presence is verified in the participant list at the top of the WhatsApp screenshot, as well as at least one unidentified staff member whose smartphone was the source of the leaked screenshot.

The group's chat logs reveal active coordination of parking exemptions and awareness of the manipulated pricing algorithm, consistent with Miguel's overhearing of Duarte and Alcibíades discussing how to "pull some strings" to avoid paying for parking in the company kitchen. Their participation indicates complicity in benefiting from, or concealing, fraudulent system operations.

Recommendations:

Based on the evidence collected, the investigation should continue by focusing on all identified actors. Particular attention should be given to Duarte Calado, Adelino Estoiro, Alcibíades Mourato, Eurico Trocos, Cremilde Trocos, and other participants in the "*Parkinator*" WhatsApp group, with warrants obtained as necessary to examine their computers, corporate systems, and communications. Forensic acquisition of digital evidence should be conducted using bitwise disk imaging, write-blockers, and other established methods to ensure integrity and prevent alteration of original data. All documents and emails should be verified for authenticity and integrity, including validation of the repository from which Catarina allegedly extracted the *diff.txt* file. Additionally, Miguel Pontes and Catarina Pato should be interviewed again to clarify their respective roles and actions.

Additionally, encrypted files suspected to contain critical data, such as the invoice tied to Futuro Inteligente (*Invoice.pdf*) and the MEMEL financial report (*FinancialReport.pdf*), should be carefully decrypted, as these files are very likely the materials referenced by Miguel in the IRC logs around the time of the phishing operation: *"First, I pulled an invoice tied to Futuro Inteligente, with a lot of suspicious transactions. And best of all, a financial report of MEMEL!!!".*

Finally, the investigation should include a thorough financial forensics review, auditing payments made to Futuro Inteligente Lda. and comparing them against contracts and actual deliverables, while investigating the flow of funds to identify any other potential hidden beneficiaries. Simultaneously, incident response measures should be implemented to ensure that any future changes to the MEMEL system are continuously monitored, preventing further exploitation or manipulation while the investigation is ongoing.